Algorithmic Foundations and Protocol Invariants for the Avalanche Matchmaking Protocol (AMP v2)

Quorum Intersection Proofs, State-Exhaustive Value Conservation, and Game-Theoretic Settlement Analysis

DRAFT — internal spec snapshot. Not for external distribution.

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Abstract

This paper establishes the mathematical specification and algorithmic invariants for the Avalanche Matchmaking Protocol (AMP v2). We analyze the safety, economic finality, and queue dynamics of decentralized, oracle-free N-player match settlement. Specifically, we prove: (1) Byzantine fault-tolerant K-of-N quorum intersection under non-equivocation assumptions across all residue classes ($N \geq 3f+1$); (2) state-exhaustive integer conservation of pooled stakes down to 1 wei across all terminal match states (`Settled`, `TimeoutClaim`, `FundingExpired`, `SilenceRefund`, `DisputeResolved`); (3) permutation invariance of multi-opponent Glicko-2 field updates over $\mathbb{R}$ with a canonically ordered computational model; (4) game-theoretic dominance of dual-deposit reporting bonds in pivotal dropout scenarios; and (5) conditional uniformity bounds of candidate lobby assignment.

Keywords: Byzantine Quorum Systems, Mechanism Design, Decentralized Matchmaking, Glicko-2, Value Conservation, Avalanche L1 Subnets.

1. Byzantine Quorum Intersection

Let an N-player match session be represented by participants $\mathcal{P} = \mathcal{H} \cup \mathcal{B}$, where $\mathcal{H}$ denotes honest participants, $\mathcal{B}$ denotes Byzantine or colluding participants, and $|\mathcal{H} \cap \mathcal{B}| = 0$. We bound the Byzantine population by $|\mathcal{B}| \leq f$.

Definition 1 (Quorum Threshold).
For a lobby of size N, canonical ladder settlement requires a concordant quorum of K participant signatures:

$$K = \lfloor 2N/3 \rfloor + 1, \quad K = N \text{ for } N \leq 3$$

Each participant holds an ECDSA keypair $(\mathrm{sk}_i, \mathrm{pk}_i)$. An attestation commits to the EIP-712 session digest over the structured message

$$m = \mathrm{keccak256}\big(\mathrm{matchId} \parallel \mathrm{chainId} \parallel \mathrm{sessionNonce} \parallel \vec{R} \parallel H_{\mathrm{transcript}}\big)$$

where $\vec{R}$ is the ranked placement vector and $H_{\mathrm{transcript}}$ is the deterministic execution trace hash.

Theorem 1 (Quorum Non-Equivocation).
For any $N \geq 3f+1$, two conflicting match states $m \neq m'$ cannot both achieve a valid quorum of size K without at least one Byzantine participant producing two distinct signatures.

Proof. Let $Q_1, Q_2 \subseteq \mathcal{P}$ be two valid quorums supporting conflicting states $m \neq m'$, with $|Q_1| \geq K$ and $|Q_2| \geq K$. By inclusion-exclusion:

$$|Q_1 \cap Q_2| = |Q_1| + |Q_2| - |Q_1 \cup Q_2| \geq 2K - N$$

We evaluate $2K - N$ across all residue classes $N = 3f + r$ where $r \in \{1, 2, 3\}$:

- **Case 1** ($r = 1, N = 3f+1$): $K = \lfloor (6f+2)/3 \rfloor + 1 = 2f+1$, hence

$$|Q_1 \cap Q_2| \geq 2(2f+1) - (3f+1) = f+1.$$

- **Case 2** ($r = 2, N = 3f+2$): $K = \lfloor (6f+4)/3 \rfloor + 1 = 2f+2$, hence

$$|Q_1 \cap Q_2| \geq 2(2f+2) - (3f+2) = f+2 \geq f+1.$$

- **Case 3** ($r = 3, N = 3f+3$): $K = \lfloor (6f+6)/3 \rfloor + 1 = 2f+3$, hence

$$|Q_1 \cap Q_2| \geq 2(2f+3) - (3f+3) = f+3 \geq f+1.$$

Thus, $|Q_1 \cap Q_2| \geq f+1$ holds for all $N \geq 3f+1$. Because the total Byzantine population is bounded by $|\mathcal{B}| \leq f$, the intersection contains at least one honest participant:

$$|(Q_1 \cap Q_2) \cap \mathcal{H}| \geq (f+1) - f = 1$$

Let $p^* \in (Q_1 \cap Q_2) \cap \mathcal{H}$. By definition of an honest node, p^* signs at most one digest per session nonce:

$$\sigma_{p^*}(m) \Rightarrow \neg\exists \sigma_{p^*}(m') \quad \text{for } m' \neq m$$

Therefore, two conflicting quorums cannot both exist unless at least one Byzantine node $p \in \mathcal{B}$ equivocally signs both m and m'. Submitting both signatures on-chain constitutes non-repudiable proof of fraud, slashing p. ∎

Remark (Liveness Bound & Threat Boundary).
At $N = 3f+1$, honest participant count $|\mathcal{H}| = 2f+1 = K$. A single honest crash-fault ($|\mathcal{H}| - 1$) starves immediate concordant quorum, triggering the optimistic grace timeout.

Furthermore, Theorem 1 guarantees **consensus non-equivocation**, not game correctness: if K Byzantine players collude to sign a fraudulent ladder with matching $H_{\mathrm{transcript}}$, the contract accepts it unless an honest minority escalates inputs to the optimistic dispute challenge window.

2. State-Exhaustive Value Conservation

Match escrow must satisfy exact mathematical conservation across all terminal execution states to prevent protocol insolvency.

Definition 2 (Dual-Deposit Escrow and Per-State Inflow).
Every participant $i \in \mathcal{P}$ deposits stake S and reporting bond B_{rep}, so $D_i = S + B_{\mathrm{rep}}$ and the deposit base is

$$\mathcal{E}_{\mathrm{deposits}} = \sum_{i=1}^{N} D_i = NS + NB_{\mathrm{rep}}.$$

If a dispute is opened, each challenging faction additionally escrows a challenge stake C_c at challenge time; the **inflow to a terminal state** s is the value actually escrowed entering that state:

$$\mathcal{E}_{\mathrm{in}}(s) = \begin{cases} NS + NB_{\mathrm{rep}} + \sum_c C_c & \text{if a dispute was opened} \\ NS + NB_{\mathrm{rep}} & \text{otherwise.} \end{cases}$$

Conservation claims are always relative to $\mathcal{E}_{\mathrm{in}}(s)$ — never to a fixed constant.

Let $\Omega_{\mathrm{gross}} = NS$. Gross protocol fee at basis points $\beta_{\mathrm{rake}} \in [0, 10000]$ is:

$$F_{\mathrm{gross}} = \left\lfloor \frac{\Omega_{\mathrm{gross}} \beta_{\mathrm{rake}}}{10000} \right\rfloor, \quad \Omega_{\mathrm{net}} = \Omega_{\mathrm{gross}} - F_{\mathrm{gross}}$$

$$F_{\mathrm{studio}} = \left\lfloor \frac{F_{\mathrm{gross}} \beta_{\mathrm{studio}}}{10000} \right\rfloor, \quad F_{\mathrm{protocol}} = F_{\mathrm{gross}} - F_{\mathrm{studio}}$$

Protocol Invariant 1 (Zero-Loss Division Remainder).
For T prize tiers ($\{t_1, ..., t_T\}$, $\sum t_j = 10000$), preliminary payouts $P_j = \left\lfloor \frac{\Omega_{\text{net}} \cdot t_j}{10000} \right\rfloor$ leave remainder $\rho = \Omega_{\text{net}} - \sum_{j=1}^{T} P_j$. By construction:

$$0 \leq \rho < T \text{ wei}$$

The implementation routes ρ to the protocol reserve as a fee, so

$$\sum_{j=1}^{T} P_j + \rho = \Omega_{\text{net}}$$

holds exactly and no value is stranded.

Theorem 2 (State-Exhaustive Conservation).
In every terminal match state $s \in \mathcal{T}_{\text{states}} = \{\text{Settled}, \text{TimeoutClaim}, \text{FundingExpired}, \text{SilenceRefund}, \text{DisputeResolved}\}$, *total ledger credits equal the per-state inflow:*

$$\sum \text{Outflows}(s) = \mathcal{E}_{\text{in}}(s) \quad \text{to 1 wei.}$$

Proof. We evaluate outflows across all five terminal states. Payout delivery is lazy (prove-your-payout claims); the identities hold for the **ledger** the moment the terminal state is recorded, since every credit below is either written at resolution or is a deterministic function of the recorded ladder and immutable tier profile.

- **State 1 (`Settled`):** Quorum $|\mathcal{S}| \geq K$ reached during the quorum window. Net pool distributes via normalized tiers (plus remainder ρ to the reserve); signers receive B_{rep} back; non-signers $\mathcal{S}$ $\mathcal{M} = \mathcal{P}$ have their bonds slashed and split 50/50 between the relayer fund and Rank 1:

$$\begin{aligned} \sum \text{Out} &= (\Omega_{\text{net}} - \rho) + \rho + F_{\text{gross}} + |\mathcal{S}| \ B_{\text{rep}} \\ &\quad + \left\lfloor \frac{|\mathcal{M}| \ B_{\text{rep}}}{2} \right\rfloor + \left(|\mathcal{M}| \ B_{\text{rep}} - \left\lfloor \frac{|\mathcal{M}| \ B_{\text{rep}}}{2} \right\rfloor \right) \\ &= NS + (|\mathcal{S}| + |\mathcal{M}|) B_{\text{rep}} = NS + NB_{\text{rep}} = \mathcal{E}_{\text{in}}(s). \end{aligned}$$

- **State 2 (`TimeoutClaim`):** The quorum window lapses; the ladder's Rank-1 participant posts a unilateral claim, unchallenged through the grace window. **Slash rule (as shipped):** every participant *except the claimant* forfeits their bond — the claimant's signature is the only one recorded in this state, so "all bonds but the claimant's" is exactly the non-signer set of the recorded settlement. The claimant recovers their own bond plus the winners' half of the $N - 1$ slashed bonds (the relayer takes the other half):

$$\begin{aligned} \sum \text{Out} &= \Omega_{\text{net}} + F_{\text{gross}} + B_{\text{rep}} + (N-1) B_{\text{rep}} \\ &= NS + NB_{\text{rep}} = \mathcal{E}_{\text{in}}(s). \end{aligned}$$

- **State 3 (`FundingExpired`):** The lobby fails to fill before the escrow deadline; only the set J of joined participants ($|J| < N$) ever deposited, so the per-state inflow is $\mathcal{E}_{\text{in}}(s) = |J| \ (S + B_{\text{rep}})$. Each joined participant claims a full refund:

$$\sum \text{Out} = \sum_{j \in J} (S + B_{\text{rep}}) = |J| \ (S + B_{\text{rep}}) = \mathcal{E}_{\text{in}}(s).$$

- **State 4 (`SilenceRefund`):** The lobby filled but the quorum and grace windows lapse with **no** claim at all (mass disconnect). No rake, no slash; every deposit refunds:

$$\sum \text{Out} = \sum_{i \in \mathcal{P}} (S + B_{\text{rep}}) = NS + NB_{\text{rep}} = \mathcal{E}_{\text{in}}(s).$$

- **State 5 (`DisputeResolved`):** A ≥ 2-signer concordant challenge contested the grace claim; the bonded verifier ruled, and the challenge stakes $\sum_c C_c$ are part of the inflow per Definition 2. Disputes take no rake; the valid ladder's tiers are paid over the gross stake pool NS (losers' stakes are thereby consumed by the ladder itself); non-losers recover bonds; the losers' bonds plus all challenge stakes form the slash pool, split 70% to the winning faction and 30% to the protocol reserve (rounding dust to the reserve):

$$\begin{aligned} \sum \text{Out} &= \underbrace{NS}_{\text{tiers over gross}} + \underbrace{(N - \ell) B_{\text{rep}}}_{\text{non-loser bonds}} \\ &\quad + \underbrace{\left(\ell B_{\text{rep}} + \sum_c C_c \right)}_{\text{slash pool, fully split}} \\ &= NS + NB_{\text{rep}} + \sum_c C_c = \mathcal{E}_{\text{in}}(s), \end{aligned}$$

  where ℓ is the losing faction's size.

Conservation holds to 1 wei across all terminal states, relative to the value actually escrowed in each. ∎

3. Multi-Opponent Rating Invariance

To update skill ratings in fields of N players, `amp-match-core` executes `glicko2_update_vs_many` across all $N - 1$ opponents within a single rating period.

Definition 3 (Field Outcomes & Glicko-2 Formulation).
For player i facing opponent $j \neq i$, outcome score s_{ij} is:

$$s_{ij} = \begin{cases} 1.0 \text{ if } \text{rank}(i) < \text{rank}(j) \\ 0.5 \text{ if } \text{rank}(i) = \text{rank}(j) \\ 0.0 \text{ if } \text{rank}(i) > \text{rank}(j) \end{cases}$$

With scale transformations $\mu = (r - 1500)/173.7178$ and $\varphi = \text{RD}\ /173.7178$ (where RD is the rating deviation), let $E_{ij} = \left(1 + \exp\left(-g(\varphi_j)(\mu_i - \mu_j)\right)\right)^{-1}$ and $g(\varphi) = \left(1 + 3\varphi^2/\pi^2\right)^{-1/2}$:

$$v_i = \left(\sum_{j \neq i} g(\varphi_j)^2 E_{ij} (1 - E_{ij}) \right)^{-1}, \quad \Delta_i = v_i \sum_{j \neq i} g(\varphi_j)(s_{ij} - E_{ij})$$

Theorem 3 (Permutation Invariance over $\mathbb{R}$).
Let $\pi : \{1, ..., N-1\} \to \{1, ..., N-1\}$ *be any permutation of opponent processing order. The mathematical tuple* $(\mu_{i'}, \varphi_{i'}, \sigma_{i'}) \in \mathbb{R}^3$ *is invariant under* π.

Proof. Functions $\psi(i,j) = g(\varphi_j)^2 E_{ij}(1 - E_{ij})$ and $\omega(i,j) = g(\varphi_j)(s_{ij} - E_{ij})$ depend strictly on the unordered pair $\{i, j\}$. Over the reals, addition is commutative and associative:

$$\sum_{k=1}^{N-1} \psi(i, \pi(k)) = \sum_{j \neq i} \psi(i,j), \quad \sum_{k=1}^{N-1} \omega(i, \pi(k)) = \sum_{j \neq i} \omega(i,j)$$

Thus, scalar field parameters v_i and Δ_i are invariant under π. Volatility $\sigma_{i'}$ is the unique zero of the objective function:

$$f(x) = \frac{e^x(\Delta_i^2 - \varphi_i^2 - v_i - e^x)}{2(\varphi_i^2 + v_i + e^x)^2} - \frac{x - \ln \sigma_i^2}{\tau^2}$$

Because $f(x)$ is parameterized entirely by invariant scalars, its root x^* is invariant under π. Derived quantities $\varphi_{i'}$ and $\mu_{i'} = \mu_i + (\varphi_{i'})^2 \sum \omega(i,j)$ are identical over $\mathbb{R}$. ∎

Remark (Computational Model & Modeling Assumptions).
Floating-point addition (IEEE-754) is non-associative. To guarantee bit-identical updates across different node architectures, `amp-match-core` enforces a **canonical accumulation order**: opponent contributions are sorted by the opponent tuple's IEEE-754 bit patterns (rating, then deviation, then score) before summation, making the floating- point sum a deterministic function of the opponent **multiset**. This property is fuzz-gated: any permutation of an opponent field must yield bit-identical $(\mu', \varphi', \sigma')$.

Modeling Note: Treating an N-player ranking as $N - 1$ independent pairwise matches is a standard Glickman rating-period approximation. Because tournament ranks are dependent, this models directional skill delta accurately but contracts φ faster than a joint multi-variate model.

4. Game-Theoretic Settlement Analysis

We evaluate the strategic incentives of an eliminated player k who placed outside the payout tiers ($r_k > T$) and considers withholding their signature to cause a match timeout.

Definition 4 (Pivotal Dropout Game Parameters).
Let $S > 0$ be the match stake, $B_{\text{rep}} > 0$ be the reporting bond, and $C_{\text{sign}} \approx 0$ be client signing gas. A player is **pivotal** if their refusal to sign reduces total signatures below quorum ($|\mathcal{S}| = K - 1$).

Theorem 4 (Dominance of Cooperation under Timeout Slashing).
Under the protocol's ***TimeoutSlash*** *rule, signing the exit attestation strictly dominates defecting whenever player k's cooperation is pivotal or quorum would otherwise be reached, and weakly dominates in the residual case where quorum fails regardless of k's action.*

Proof. Under protocol rules, a match timeout does **not** refund stake S to uncooperative participants. If quorum fails:

1. The match enters the `TimeoutClaim` state post-grace.
2. The highest-attested valid survivor (Rank 1) claims net escrow Ω_{net}.
3. Per the State-2 slash rule, **every participant except the claimant** forfeits B_{rep} — including any player who signed but whose signature did not enter a recorded settlement. (Equivalently: bond recovery requires being the recorded signer of a **settling** ladder; a signature that never settles recovers nothing.)

We evaluate player $k \neq$ claimant's terminal payoff matrix:

- **Scenario A (quorum reached without k, $|\mathcal{S} \setminus \{k\}| \geq K$):**

$$U_k(\text{Sign}) - U_k(\text{Defect}) = \left(-S + B_{\text{rep}} - C_{\text{sign}}\right) - \left(-S - B_{\text{rep}}\right) = 2B_{\text{rep}} - C_{\text{sign}} > 0.$$

- **Scenario B (k strictly pivotal, $|\mathcal{S} \setminus \{k\}| = K - 1$):** If k cooperates the ladder settles with k in the signer set:

$$U_k(\text{Sign}) = -S + B_{\text{rep}} - C_{\text{sign}}.$$

  If k defects, quorum starves, the match settles via `TimeoutClaim`, and k (a non-claimant) is slashed:

$$U_k(\text{Defect}) = -S - B_{\text{rep}}. \quad U_k(\text{Sign}) - U_k(\text{Defect}) = 2B_{\text{rep}} - C_{\text{sign}} > 0.$$

- **Scenario C (quorum fails regardless of k):** `TimeoutClaim` slashes every non-claimant whether or not they signed, so $U_k(\text{Sign}) = U_k(\text{Defect}) = -S - B_{\text{rep}}$ and signing is weakly dominant (the $C_{\text{sign}} \approx 0$ gap is one-directional only through the claimant's tip, which k forgoes either way).

Because defection never triggers a stake refund and always forfeits B_{rep} whenever it matters, cooperation dominates defection across all game states. ■

Remark (Scope of Theorem 4).
Theorem 4 is a corollary of the State-2 slash rule, not the protocol's economic centerpiece: it states that a reporting bond of any positive size makes exit-signing incentive-compatible for losing, non-pivotal players. It does not model collusion among $\geq K$ signers (covered by Theorem 1's fraud evidence) or verifier capture in disputes (covered by the bonded-verifier stake).

5. Queue Allocation & Entropy Bounds

To prevent pre-coordinated collusion in open staked FFA queues, AMP partitions the matchmaking queue using a commit-reveal shuffle.

Definition 5 (Commit-Reveal Shuffle).
Candidates submit blinded commitments $H_i = \text{keccak256}(p_i \parallel S \parallel \text{salt}_i)$. At candidate pool size $M \geq N$, the coordinator commits to target block height $H_{\text{target}} = H_{\text{current}} + \Delta$ ($\Delta \geq 2$). After salts are revealed, the assignment seed is

$$\Xi = \text{keccak256}\left(\mathcal{B}_{\text{target}} \parallel \text{salt}_1 \parallel \text{salt}_2 \parallel \ldots \parallel \text{salt}_R\right)$$

over the R revealed salts, where $\mathcal{B}_{\text{target}}$ is the Avalanche blockhash at H_{target}.

Theorem 5 (Conditional Allocation Uniformity).
Conditioned on a seed Ξ drawn uniformly, $\Xi \sim \text{Uniform}(\{0,1\}^{256})$, a deterministic Fisher-Yates shuffle achieves uniform distribution over candidate permutations. For any two candidates $i \neq j$, lobby co-location probability is:

$$\mathbb{P}(\text{Lobby}(i) = \text{Lobby}(j)) = \frac{\binom{M-2}{N-2}}{\binom{M-1}{N-1}} = \frac{N-1}{M-1}$$

Proof. Under a uniform random permutation of M candidates into a lobby of size N, the number of subsets of size N containing both i and j is $\binom{M-2}{N-2}$. Total possible size-N lobbies from pool M is $\binom{M-1}{N-1}$ (fixing candidate i, choose the remaining $N-1$ from $M-1$). The ratio yields $\frac{N-1}{M-1}$.

For a colluding cartel of size c, co-locating all c members in a single lobby decays as:

$$\mathbb{P}(\text{Cartel Co-Location}) = \prod_{k=1}^{c-1} \frac{N-k}{M-k} \in \mathcal{O}\left(\left(\frac{N}{M}\right)^{c-1}\right)$$

■

Remark (Proposer Discretion & Threat Boundaries).
Avalanche Snow consensus provides sub-second finality, but single-block proposers retain transaction-ordering and block-withholding discretion. True unbiasability of $\mathcal{B}_{\text{target}}$ is therefore an engineering approximation: Theorem 5 is **conditional** on the seed's uniformity, which is assumed, not proven.

Mitigations: (1) Reveal deadlines slash deposits of unrevealed commitments, preventing selective aborts; (2) The production roadmap transitions Ξ to threshold BLS beacons via Avalanche Warp Messaging (AWM).

6. Machine-Checked Invariants & Fuzz Gates

Protocol invariants are continuously asserted via stateful property testing and fuzzing:

Protocol Invariant	Formal Bound	Automated Test Gate
Quorum Non-Equiv.	$\lvert Q_1 \cap Q_2 \rvert \geq f + 1$	Conflicting-quorum terminal test
State Conservation	$\sum \text{Out} = \mathcal{E}_{\text{in}}(s)$ to 1 wei	Foundry invariant fuzz (256 runs x 128k calls)
Glicko π-Invariance	Bit-identical $(\mu', \varphi', \sigma')$	`proptest` permutation suite (10k cases)
Pull Accounting	Zero external calls in resolution	Reverting-receiver isolation test
Shuffle Uniformity	Lemire-debiased bounded draws	Distribution spread test (7k draws/bucket)

These gates **test** the invariants; they are not proofs of Theorems 2 or 4. The conservation fuzz enforces the Theorem-2 identities mechanically across randomized op sequences; the quorum and payoff identities are proven above.

7. Gas Execution Bounds

On-chain verification in `AMPMultiplayer.sol` records the settlement (constant storage) and verifies K packed signatures; per-player payouts are claimed lazily, so settlement gas scales with signatures, not recipients:

Lobby (N)	Quorum (K)	Measured settlement gas
8 players	6 sigs	152,962 gas (spec gate: under 250k)
16 players	11 sigs	195,844 gas
64 (BR)	43 sigs	404,021 gas

By restricting balance distribution to internal ledger credit (`claimable[player] += reward`), 64-player Battle Royale settlements avoid iterative external call overhead, remaining well within Avalanche block gas limits (30,000,000 gas).

8. Deployments and Verification Status

All contract deployments live on the Avalanche **Fuji testnet**. Nothing is deployed to mainnet.

- **v1 1v1 settlement stack (live on Fuji, Sourcify exact-match verified):** `AMPRegistry` at `0xf6B0eA6c88c574c4BbEAdC186AAfe72C43C2cDc2` and `AMPSettlement` at `0x78ec93e66255a74873d20DD62C6595A389272126`.
- **v2 N-player engine (live on Fuji, Sourcify full-match verified):** `AMPMultiplayer` at `0xcabf7b626172fE55d54f03c346563671AbcC77f7`, branch `n-player-multiplayer`. Implements dual-deposit escrow, bitmask quorum verification, and immutable payout profiles with lazy claims.
- **Match core library (`amp-match-core`):** Pure Rust core implementing `party.rs`, team/FFA/BR queue topologies, commitments and the blockhash shuffle, and canonical Glicko-2 field updates.

9. Appendix: Theorem-to-Implementation Cross-Reference

Every theorem, state, and invariant in this document, mapped to the contract function or library routine that implements it and the automated gate that exercises it. Contract references are to `AMPMultiplayer.sol` unless prefixed; library references are to `amp-match-core` (Rust).

Claim (this doc)	Implementation site	Automated gate
Def. 1 — quorum threshold K	`quorumOf()`	`test_LobbyFillsAndTransitionsToReady` (asserts $K = 6$ at $N = 8$)
Thm. 1 — non-equivocation	`_verifyLadderQuorum` sig recovery + terminal `State.Settled`	`test_SettlementIsTerminal` (conflicting quorum post-settle reverts); `test_MisattributedSignatureRej`
Inv. 1 — remainder ρ	`claimFees()` treasury branch: `settledNetPool – tierTotal`	`test_QuorumSettlement_Pays_Fees_Bonds` (dust to reserve, conservation exact)
Thm. 2, State 1 `Settled`	`settleMultiplayer` → `_settleAndRecord`; payouts via `claimPayout` / `claimFees`	`test_QuorumSettlement_Pays_Fees_Bonds`; `invariant_TerminalStateDrainsExactly`
Thm. 2, State 2 `TimeoutClaim`	`unilateralClaim` → `finalizeGraceWith` (claimant-only signer record)	`test_GraceClaim_Finalizes_Unchallenged` ($N - 1$ bonds slashed); `test_GraceClaim_Requ` `test_GraceWindow_ClaimBeforeQuorumLapse_`
Thm. 2, State 3 `FundingExpired`	`cancelLobby` (refunds joined set only)	`test_CancelUnfilledRefundsDeposits`
Thm. 2, State 4 `SilenceRefund`	`expireRefund`	`test_ExpireRefund_AfterTotalSilence`
Thm. 2, State 5 `DisputeResolved`	`challengeClaim` → `resolveDispute` → `_payoutDispute`	`test_Dispute_ChallengersWin_SlashClaiman` `test_Dispute_ClaimantWins_SlashChallenge` `test_Dispute_VerdictMustMatchWinnerLadde`
Thm. 2 — conservation (all states)	pull-only `claimable` ledger + `withdraw`	`invariant_BalanceCoversCredits`; `invariant_TerminalStateDrainsExactly` (256 runs × 128k calls); per-state exact asserts in each unit test above
Thm. 3 — permutation invariance	`amp-match-core: glicko2_update` (canonical bit-pattern sort)	`rating_updates_vs_many_are_permutation_invariant` (10k-case fuzz, incl. blockhash-shuffle permutation); `vs_many_is_order_`
Thm. 4 — timeout dominance	State-2 slash rule in `_settleAndRecord` (non-signer bonds 50/50)	grace tests above; payoff structure asserted in `test_GraceClaim_Finaliz`
Thm. 5 — allocation uniformity	`amp-match-core: commit.rs` `shuffle_by_blockhash` (Lemire debiasing)	`shuffle_is_a_permutation`; `below_is_within` (7k draws/bucket); `shuffle_is_deterministic`
Pull-accounting (DoS isolation)	`withdraw` isolated per player	`test_WithdrawIsolatedFromRevertingReceiv`
Gas bounds (§7)	`settleMultiplayer` (O(1) storage; lazy claims)	`test_QuorumGasBound_At_N8_Under_250k` (152,962); `test_GasProfile_N16` (195,844); `test_GasProfile_BR64` (404,021)

Contract: `contracts/src/AMPMultiplayer.sol`, deployed at the address in §8. Tests: `contracts/test/AMPMultiplayer.t.sol`, `contracts/test/AMPMultiplayer.invariants.t.sol`, `amp-match-core/tests/fuzz_gates.rs`. Gas figures are measured (optimizer_runs = 64, via_ir).

Engineering Sign-Off (draft):
The invariants specified here are implemented in `AMPMultiplayer.sol` and `amp-match-core` and exercised by 91 Foundry tests (including the two stateful conservation invariants) and 106 Rust tests (including the permutation and party-graph fuzz gates). Fuzz results are empirical evidence, not proof; the proofs in this document stand on the stated assumptions. External audit and timelock governance precede any mainnet deployment.